

Research problem, bounded gap, and proposed contribution

Research problem

Anomaly-detection deployments must balance detection performance against communication and compute costs. Edge computing is motivated in part by reduced response time and bandwidth cost through moving processing closer to data sources [1]. Edge-intelligence research treats inference placement and model selection as runtime trade-offs involving accuracy, latency, computation, memory, energy, and communication [2]. The project proposes to compare one centralized cloud route, one fixed-model edge route, and a resource-aware adaptive edge route.

Existing approaches and limitations

Published work has established multiple relevant components, but these should not be conflated with evidence for this project. Isolation Forest is an established isolation-based anomaly detector designed with computational and memory efficiency in mind [3]. LSTM encoder-decoder approaches have been used for reconstruction-based time-series anomaly detection [4], and LSTM telemetry work has explicitly addressed dynamic thresholding and false-positive mitigation [5]. Resource-aware adaptive selection among pretrained models has also been studied for embedded inference, albeit in image classification rather than this project's stated anomaly-detection comparison [6].

These sources do not establish a universal edge advantage, nor do they establish that model adaptation will improve detection. Local processing can avoid sending all raw observations, but it is constrained by local compute resources; cloud processing can access centralized compute while incurring communication and network costs [1], [2]. A meaningful comparison must therefore report detection quality and system costs together and use consistent measurement definitions.

Research gap

Within the academic records and passages retrieved for this draft, no directly comparable primary evaluation was identified that combined all of the following in one reproducible anomaly-detection experiment: (1) a centralized cloud baseline, (2) a fixed-model edge baseline, (3) resource-aware edge selection across several model classes, (4) explicit event-only communication accounting, and (5) jointly reported detection and system metrics under a common end-to-end latency boundary. This statement describes the scope and outcome of this search; it is not a claim that no such prior research exists.

More immediately, the project summary leaves an empirical verification gap. It provides aggregate reported results but not the actual dataset or generator, confusion matrices, controller trace, full evaluation definitions, system timing instrumentation, run-to-run variability, or original JSON file. As a result, the displayed comparison cannot yet establish reproducibility, statistical uncertainty, or an adaptive benefit.

Proposed approach

Retain the three architectures as stated, but run them on identical data partitions and define the same start/end timing boundaries for all latency measures. Publish model configurations and threshold selection rules; report confusion matrices, class prevalence, precision, recall, F1, ROC-AUC, PR-AUC, bytes per sample and per alert, and run-level CPU/memory/latency measurements. Log each adaptive decision, selected model, resource state, utility components, and transmission action. Test across multiple resource and network conditions and on separately identified real-world data, in addition to the reported synthetic workload.

The candidate model classes are described as Isolation Forest (LIGHT), Autoencoder (STANDARD), and LSTM Autoencoder (HEAVY). These labels and their actual resource ordering should be verified with measurements. Prior research supports isolation-based detection, LSTM sequence reconstruction, and adaptive embedded model selection as methodological antecedents, but does not validate the specific model implementations in this project [3], [4], [6].

Expected contribution

If implemented and tested as specified, the defensible contribution would be an auditable comparison of centralized, static-edge, and resource-aware adaptive-edge anomaly detection with joint reporting of predictive and systems outcomes. It could quantify when transmission reduction or local inference comes with a detection or resource cost, and whether the controller changes the trade-off beyond a fixed edge baseline. These are expected contributions, not demonstrated findings. The currently supplied data show identical reported detection measures for static and adaptive edge, so they do not yet establish an adaptive detection gain.

Research questions

1. How do the three approaches compare on threshold-dependent and threshold-independent detection metrics?
2. How do they compare on consistent end-to-end latency, network bytes, CPU, and memory across controlled resource/network conditions?
3. Does the adaptive policy improve a prespecified multi-objective utility or satisfy pre-defined constraints better than static edge processing?
4. Are the observed outcomes stable across repeated runs and on real-world data?

Hypotheses to test

- **H1:** Edge methods reduce transmitted bytes relative to centralized processing when configured to send anomaly events only.

- **H2:** Under matched, end-to-end timing, edge processing reduces latency in defined workload/network conditions.
- **H3:** Adaptive edge improves a predeclared joint objective relative to static edge while respecting minimum detection constraints.
- **H4:** Adaptive edge is non-inferior to static edge on a prespecified primary detection measure within a justified margin.

Hypotheses H1–H4 are proposals. The supplied current results do not establish them; H3 in particular requires evidence that the adaptation policy's decisions alter outcomes and improve the chosen utility.

Scope and assumptions

The current draft is limited to the reported synthetic-data experiment and the project architecture description supplied by the participant. It assumes only for presentation that the metric names have their conventional meanings and that values correspond to the three configurations listed. It does not assume the class balance, data size, hardware, network settings, hyperparameters, trial count, or statistical significance. The reported latency comparison is explicitly treated as not directly comparable because timing boundaries differ.

Limitations and future validation needs

Before making publication claims, verify the source code, configuration, data and labels, generated splits, threshold calibration, model definitions, actual adaptive actions, metric code, resource-monitoring code, and `results/final_results.json`. Recover experimental logs and repeated trials; provide confusion matrices and class support; document hardware and network conditions; run matched end-to-end timing; and validate on real data. A measured model-selection benefit may or may not emerge. If adaptation continues to select only STANDARD, frame the contribution as a documented comparison or policy feasibility assessment rather than an improvement over the static baseline.

References

- [1] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet of Things Journal*, 2016, doi: [10.1109/JIOT.2016.2579198](https://doi.org/10.1109/JIOT.2016.2579198).
- [2] Z. Zhou, X. Chen, E. Li, L. Zeng, K. Luo, and J. Zhang, "Edge intelligence: Paving the last mile of artificial intelligence with edge computing," *Proceedings of the IEEE*, June 2019, doi: [10.1109/jproc.2019.2918951](https://doi.org/10.1109/jproc.2019.2918951).
- [3] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation-based anomaly detection," *ACM Transactions on Knowledge Discovery from Data*, 2012, doi: [10.1145/2133360.2133363](https://doi.org/10.1145/2133360.2133363).
- [4] P. Malhotra, A. Ramakrishnan, G. Anand, L. Vig, P. Agarwal, and G. Shroff, "LSTM-based encoder-decoder for multi-sensor anomaly detection," *arXiv*, 2016, Available: <https://arxiv.org/pdf/1607.00148v2>

- [5] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding," 2018, doi: [10.1145/3219819.3219845](https://doi.org/10.1145/3219819.3219845).
- [6] B. Taylor, V. S. Marco, W. Wolff, Y. Elkhatib, and Z. Wang, "Adaptive deep learning model selection on embedded systems," *Lancaster EPrints (Lancaster University)*, pp. 31–43, 2018, doi: [10.1145/3211332.3211336](https://doi.org/10.1145/3211332.3211336).